

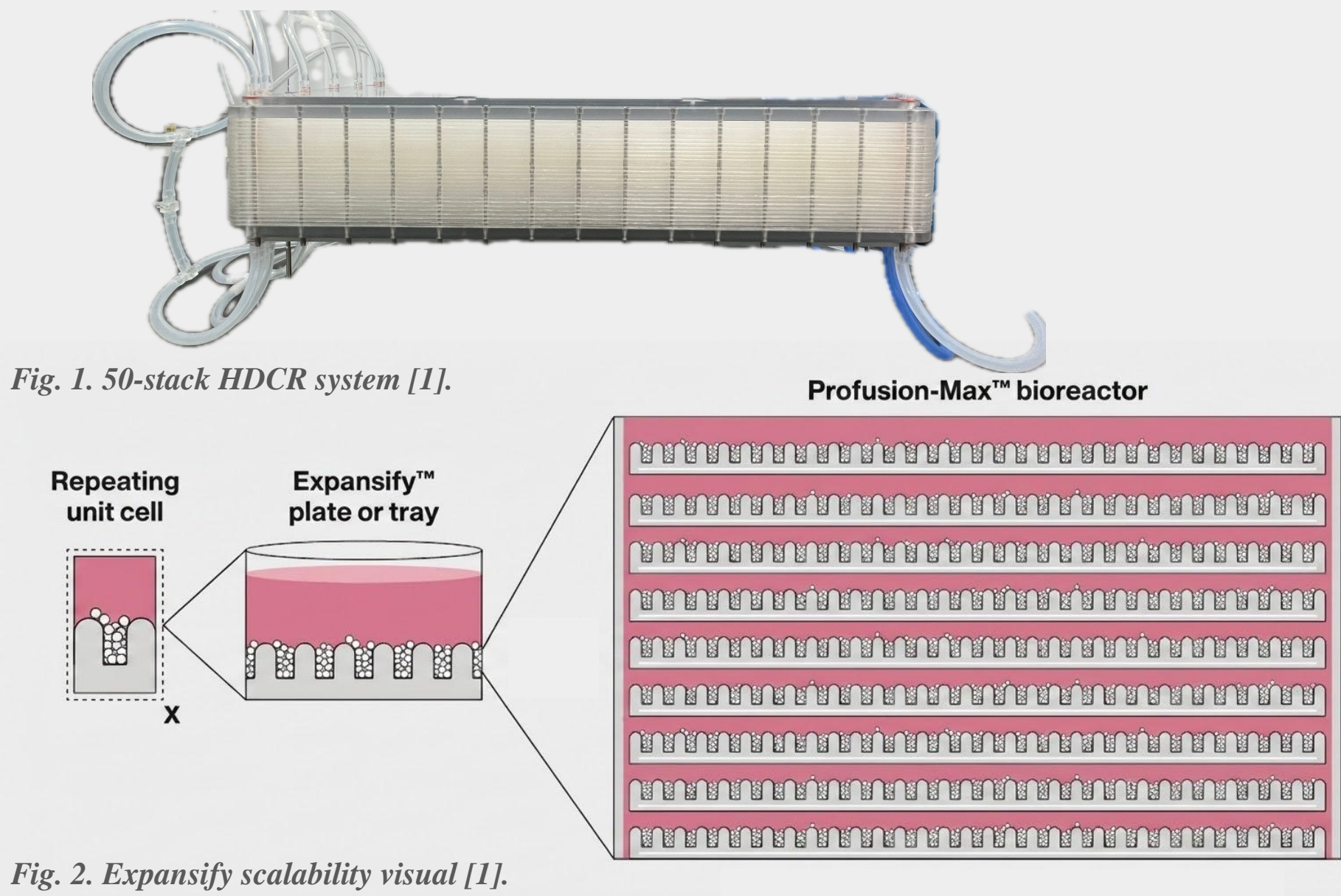


Multi-Axis Motion and Vibration System for Bioreactor Cell Seeding and Recovery

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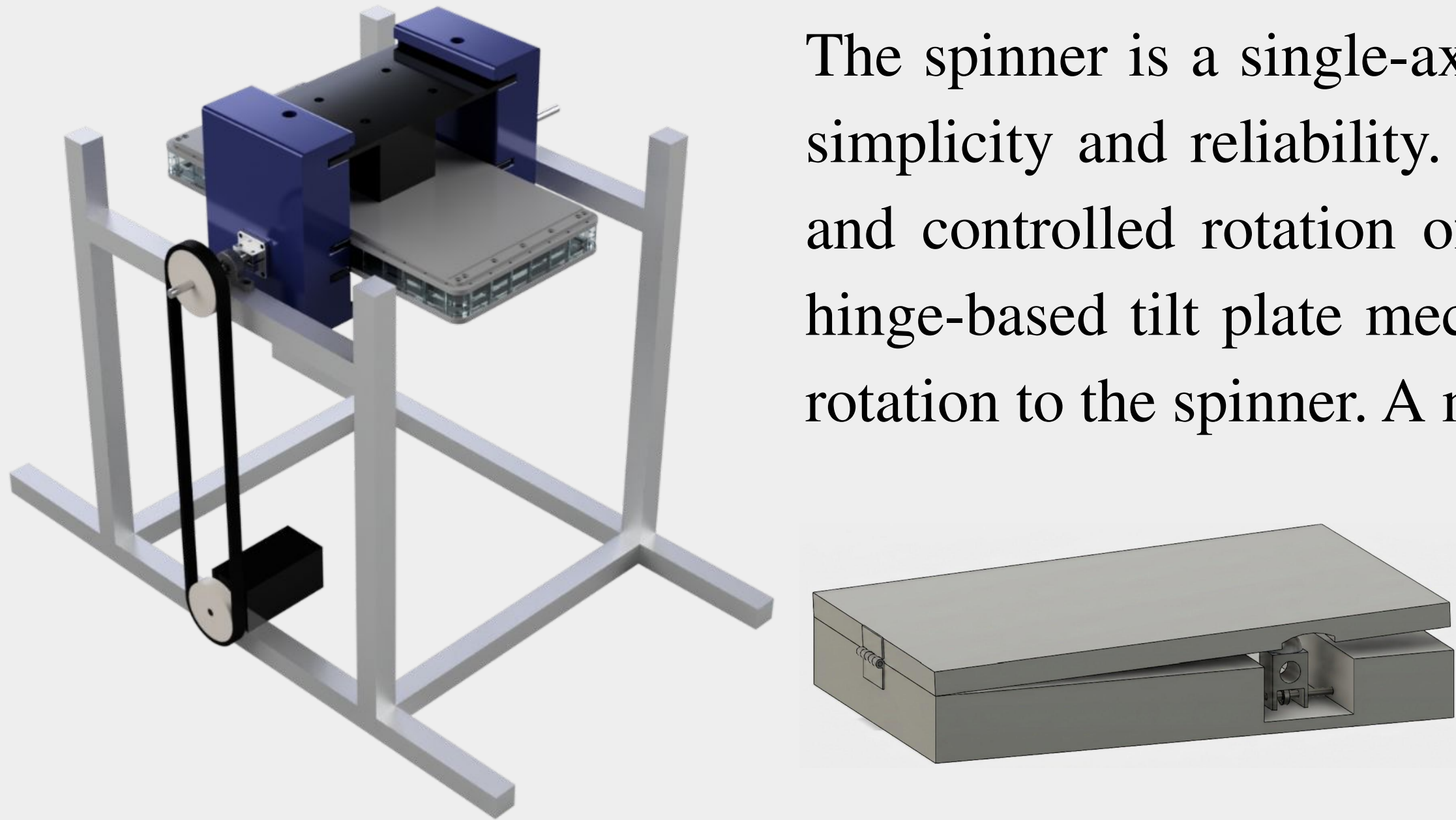


Introduction and Problem Statement



XDemics designs innovative bioreactors to increase cell culture throughput and expand access to life-saving biologics by improving the scale and efficiency of cell seeding and harvesting. Currently, a key bottleneck in the process is the manual manipulation (rotation, vibration, etc.) of cell culture containers, which prevents large-scale production and introduces user variability. **The XDemics clinic team is tasked with designing, constructing, and testing a robotic system to automate bioreactor cell seeding and recovery.**

Rotation

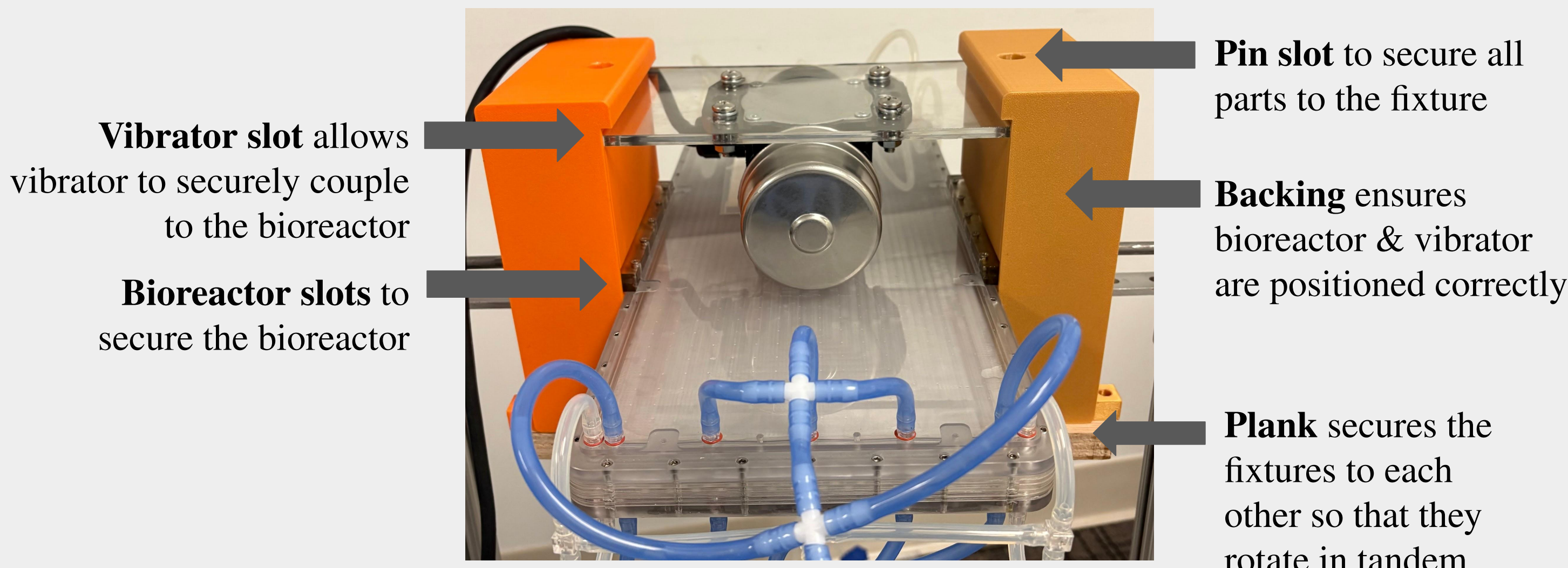


The spinner is a single-axis rotational system designed with an emphasis on simplicity and reliability. This straightforward approach provides consistent and controlled rotation of the bioreactor. The whole system will rest on a hinge-based tilt plate mechanism that was designed to add a second axis of rotation to the spinner. A motorized pusher lifts the top plate to control tilt, allowing the bioreactor to move to a 15° tilt for wetting. The figure on the left shows the full assembly, while the figure on the right shows the hinge plate mechanism.

Design Overview

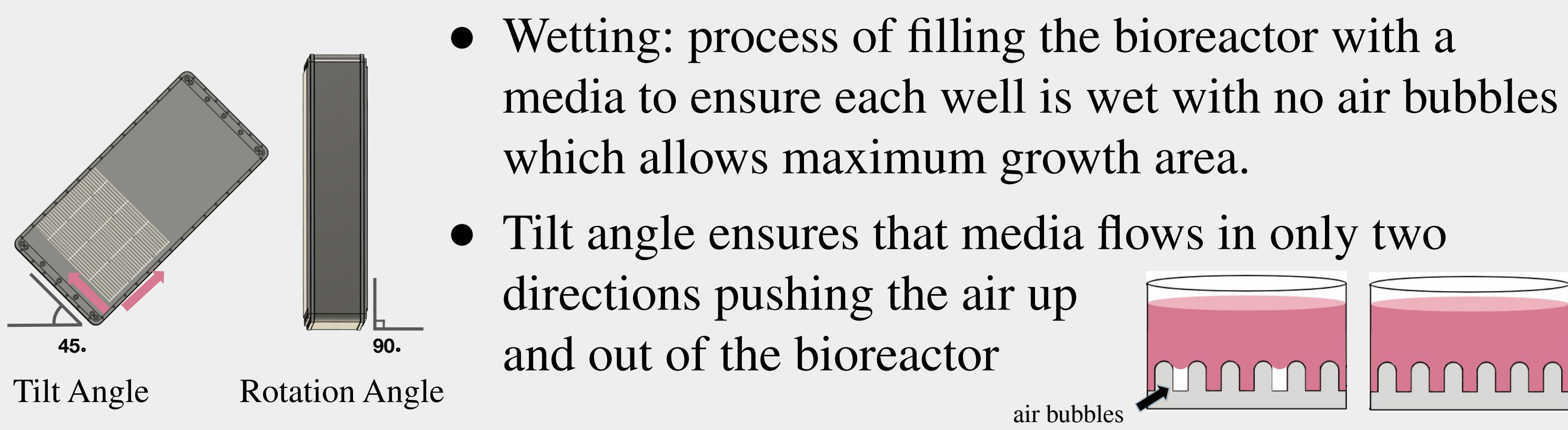
- Rotation**
 - Rotate 90° for cell seeding and harvesting
 - Rotate 180° for transfection and harvesting
 - Tilt 15° for wetting
- Fixturing**
 - Maintain fixed orientation of bioreactor
 - Support weight and loads of bioreactor
 - Allow quick installation and removal
- Agitation**
 - Impact underside of the bioreactor
 - Agitate the cells out of the wells for harvesting and transfection
- Wetting**
 - Tilt a minimum of 15° to successfully wet the bioreactor
 - Ensure all wells wet to allow maximum cell growth

Fixturing & Agitation



The fixture secures and aligns the bioreactor for controlled rotation and vibration. Slotted guides enable easy insertion and positioning, with a central pin for locking, and an upper slot for vibration coupling. The vibrator is coupled to the bioreactor for optimal vibration transfer, confirmed by acceleration measurements with a vibration meter.

Wetting



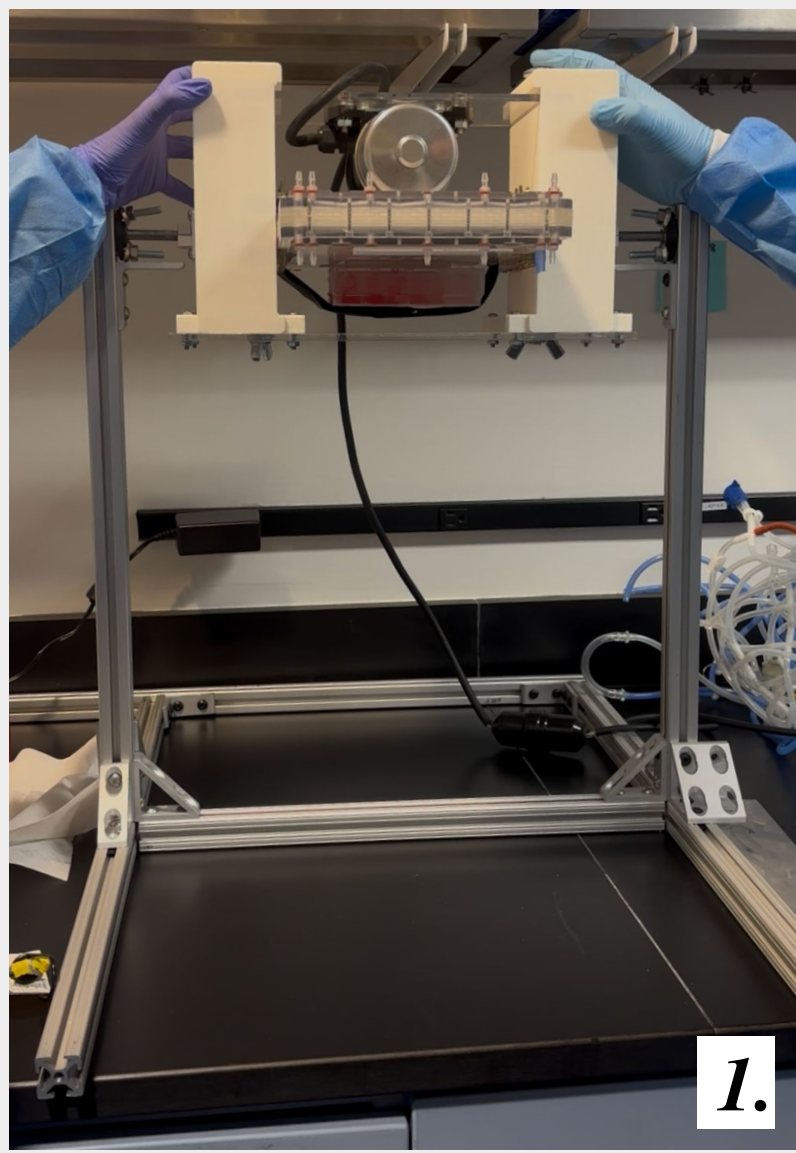
- Wetting: process of filling the bioreactor with a media to ensure each well is wet with no air bubbles which allows maximum growth area.
- Tilt angle ensures that media flows in only two directions pushing the air up and out of the bioreactor

This testing identified the minimum tilt angle needed for complete membrane wetting without trapped air by perfusing 40% ethanol through a 5-membrane HDCR system and evaluating bubble formation and fluid coverage.

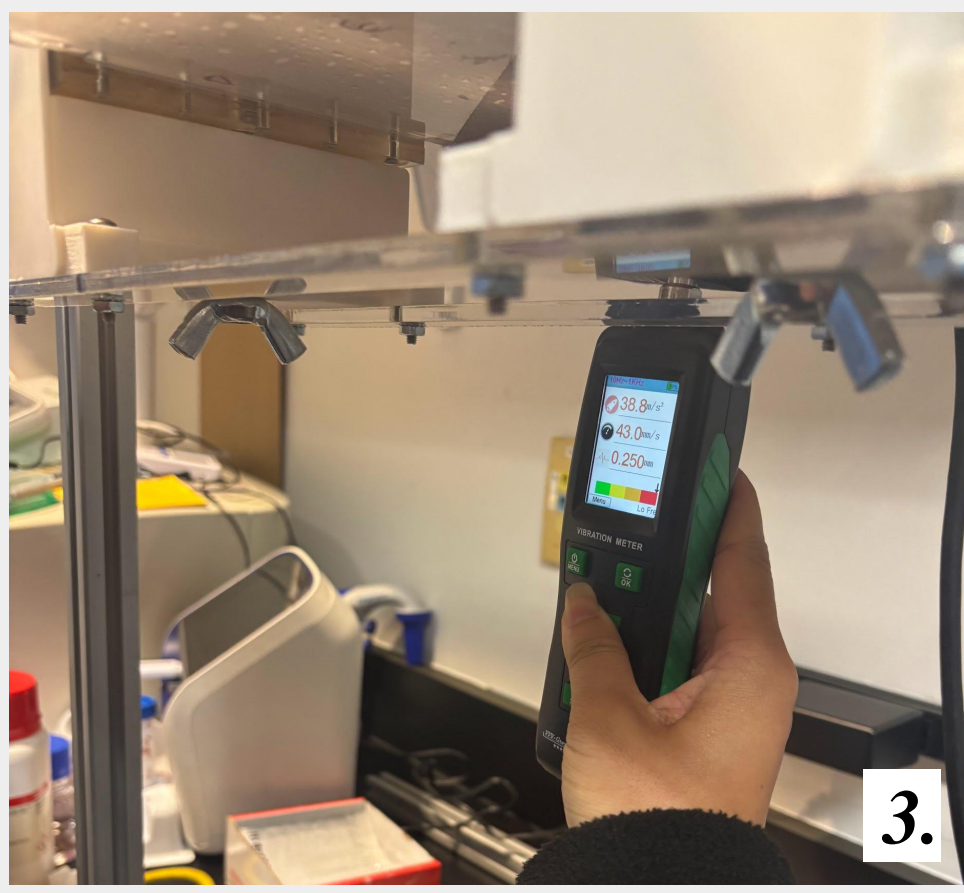
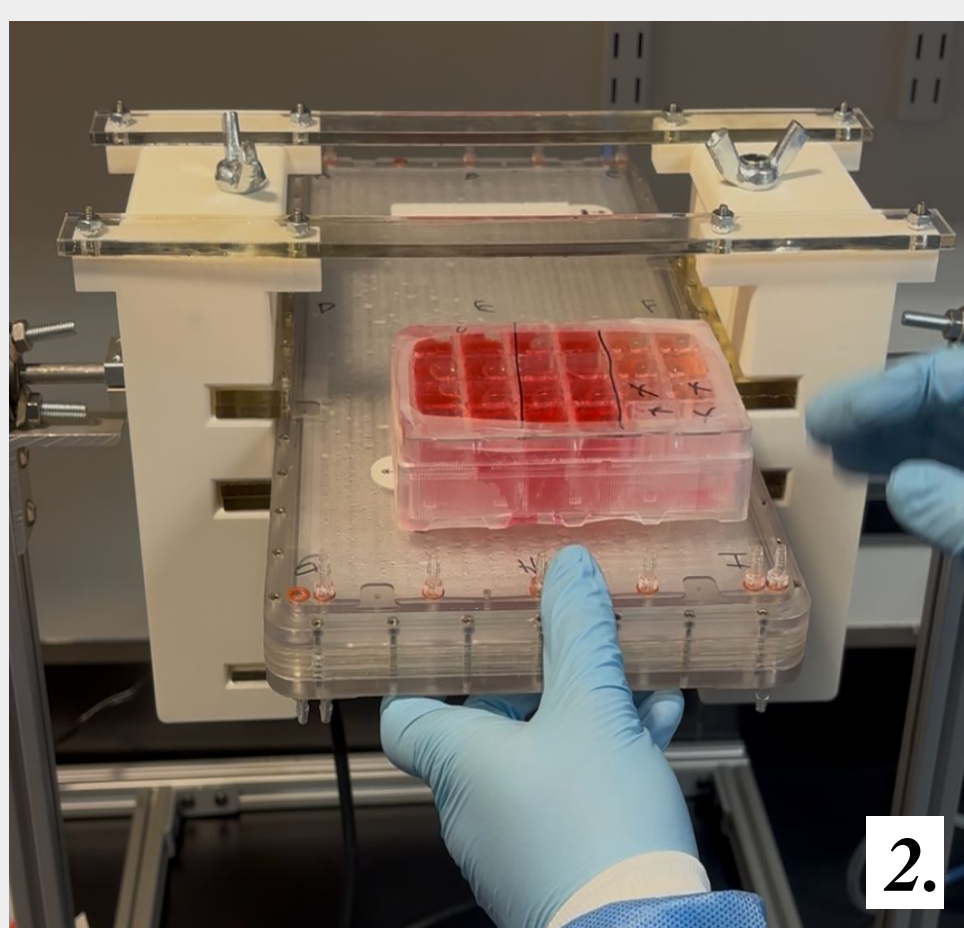
Wetting Angle Pass/Fail Table	
0°	Fail
5°	Fail
10°	Fail
10° with vibration	Fail
15°	Pass



Experimental Setup



- Full system setup showing bioreactor on stand coupled to vibrator
- 24-well plate attached to bioreactor to test vibration effect on cells
- Measuring vibration on 24-well plate with vibrometer

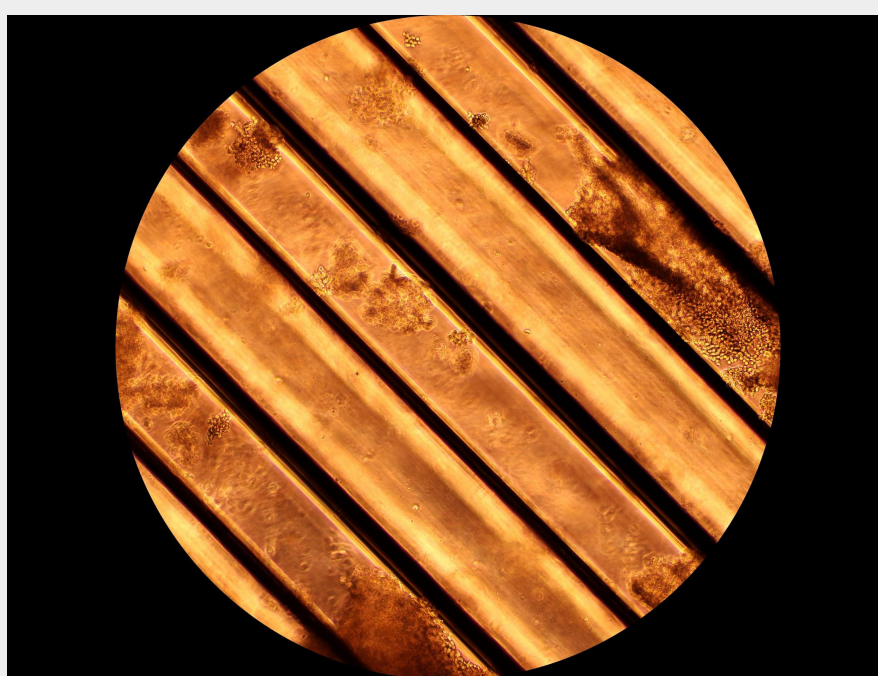


Site Visit Experimental Results

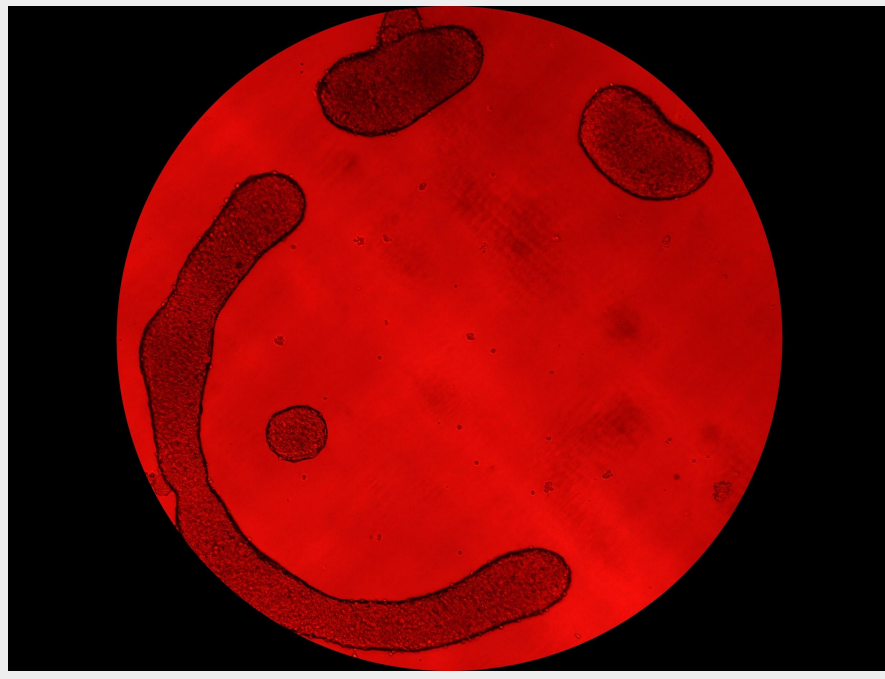
Test	Test Description	Results
1	30 second vibration, no enzyme	Adherent cells remained attached; spheroid cells were partially removed (~10–20%); suspension cells were mostly removed (>80%).
2	5 minute vibration, no enzyme	Adherent cells remained attached; most spheroid cells (~80%) and nearly all suspension cells (>95%) were removed.
3	30 second vibration after 10 min enzyme treatment	More than 80% of adherent cells detached and dropped.
4	5 minute vibration after 15 min enzyme treatment	More than 95% of adherent cells detached, and nonadherent cells were almost completely removed (~100%).
5	Incubate-shake-incubate-shake with enzyme	More than 95% of adherent cells were removed with ~84% viability, though extra pipetting was needed for single-cell suspension.



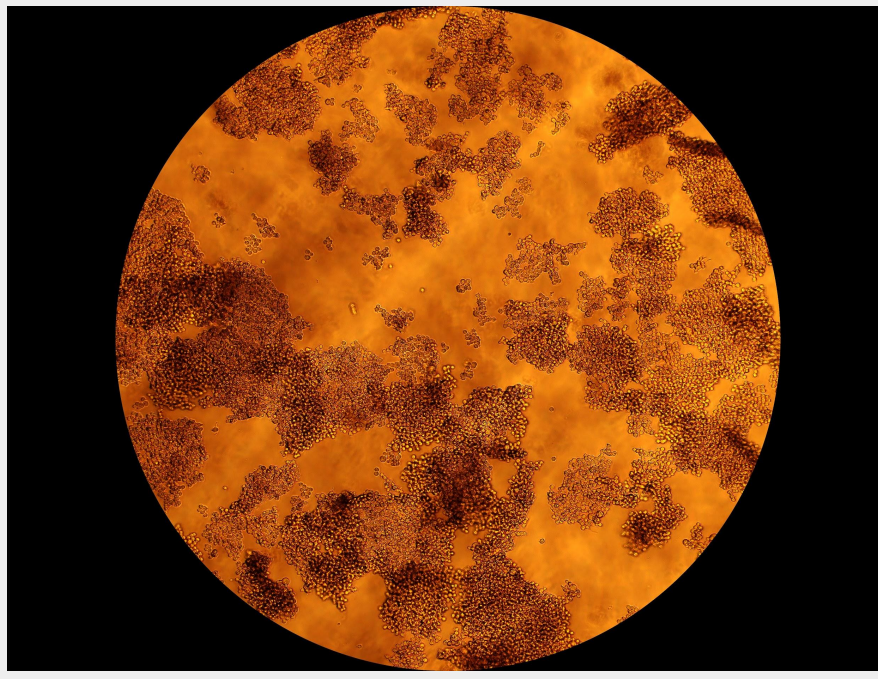
Test 3, non-adherent hek293 cells clumped but removed from wells



Test 3, adherent hek 293 cells mostly removed from wells



Test 2, non-adherent hek293 cells, heavily clumped together but removed from wells



Test 5, adherent hek293, better single cell suspension observed

Conclusion and Next Steps

The attachment, fixturing, and agitation mechanisms developed in this project successfully supported the key processes required for wetting, cell seeding, and cell recovery. The site visit demonstrated strong system performance and effective removal of cells from the bioreactor’s niches. Future work for XDemics focuses on optimizing the distance between the vibrator and bioreactor to maximize vibration transfer. Additional efforts will also include development of a motor control system to support automation, and product design to ensure conformity with cleanroom standards.

Acknowledgements and References

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[1] XDemics, “XDemics Intro_8-31-25 v2,” internal company presentation, August 2025.